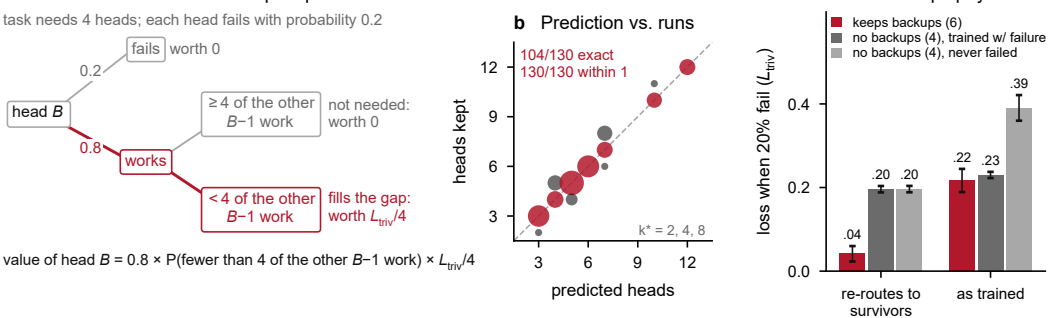
a Is one more head worth its upkeep?
task needs 4 heads; each head fails with probability 0.2
fails
worth 0
0.2
head B
0.8
works
≥ 4 of the other B−1 work
not needed: worth 0
< 4 of the other B−1 work
fills the gap: worth L_triv/4
value of head B = 0.8 × P(fewer than 4 of the other B−1 work) × L_triv/4
b Prediction vs. runs
104/130 exact
130/130 within 1
k* = 2, 4, 8
heads kept
predicted heads
12
9
6
3
3
6
9
12
c Do backups pay off?
keeps backups (6)
no backups (4), trained w/ failure
no backups (4), never failed
loss when 20% fail (L_triv)
0.0
0.2
0.4
.04
.20
.20
.22
.23
.39
re-routes to survivors
as trained